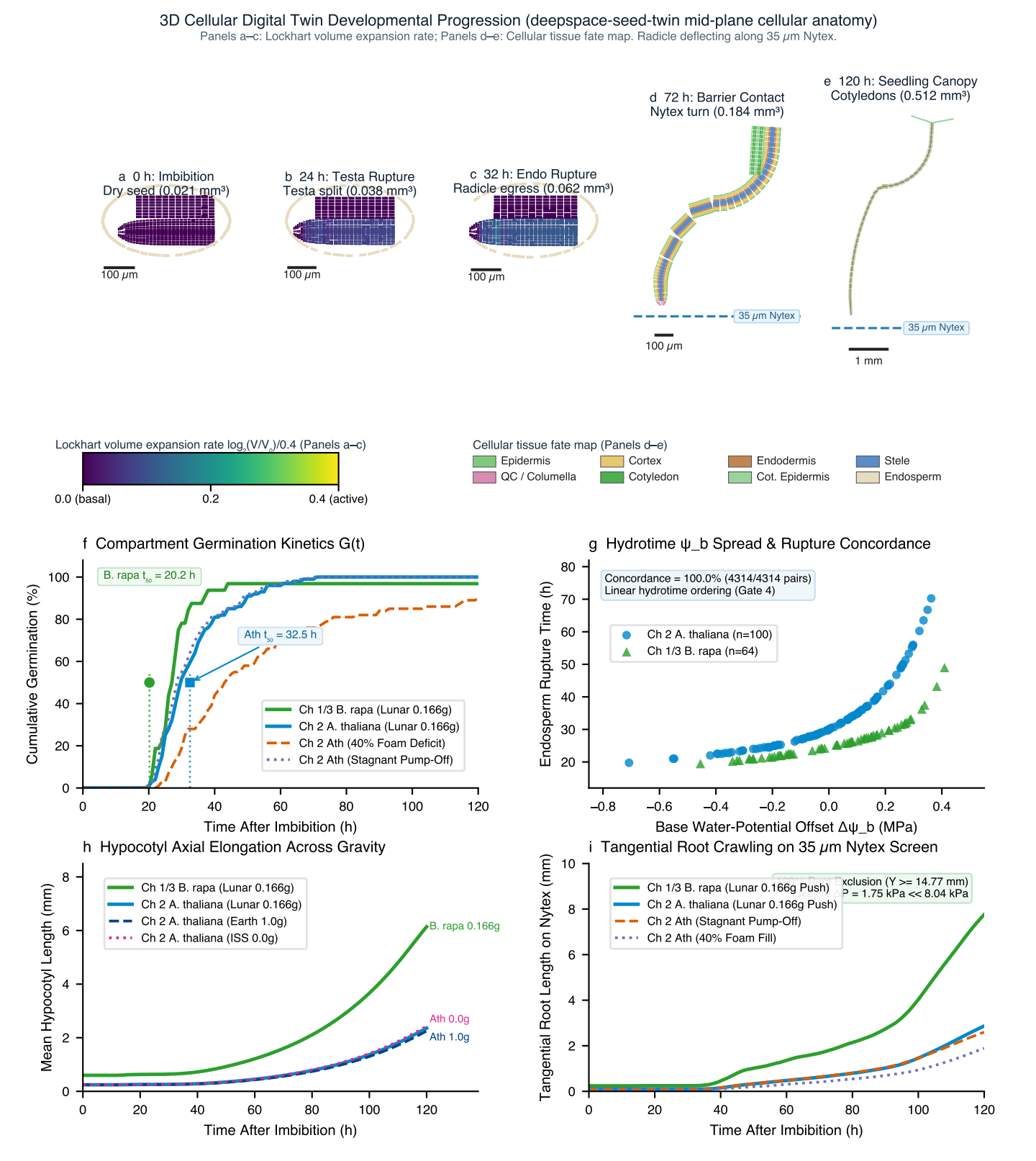

3D Cellular Digital Twin Developmental Progression (deepspace-seed-twin mid-plane cellular anatomy)
Panels a–c: Lockhart volume expansion rate; Panels d–e: Cellular tissue fate map. Radicle deflecting along 35 μm Nytex.
a 0 h: Imbibition
Dry seed (0.021 mm³)
b 24 h: Testa Rupture
Testa split (0.038 mm³)
c 32 h: Endo Rupture
Radicle egress (0.062 mm³)
d 72 h: Barrier Contact
Nytex turn (0.184 mm³)
e 120 h: Seedling Canopy
Cotyledons (0.512 mm³)
100 μm
35 μm Nytex
1 mm
Lockhart volume expansion rate log2(V/V0)/0.4 (Panels a–c)
0.0 (basal)
0.2
0.4 (active)
Cellular tissue fate map (Panels d–e)
Epidermis
Cortex
Endodermis
Stele
QC / Columella
Cotyledon
Cot. Epidermis
Endosperm
f Compartment Germination Kinetics G(t)
B. rapa t50 = 20.2 h
Ath t50 = 32.5 h
Ch 1/3 B. rapa (Lunar 0.166g)
Ch 2 A. thaliana (Lunar 0.166g)
Ch 2 Ath (40% Foam Deficit)
Ch 2 Ath (Stagnant Pump-Off)
Cumulative Germination (%)
Time After Imbibition (h)
g Hydrotime ψ_b Spread & Rupture Concordance
Concordance = 100.0% (4314/4314 pairs)
Linear hydrotime ordering (Gate 4)
Ch 2 A. thaliana (n=100)
Ch 1/3 B. rapa (n=64)
Endosperm Rupture Time (h)
Base Water-Potential Offset Δψ_b (MPa)
h Hypocotyl Axial Elongation Across Gravity
Ch 1/3 B. rapa (Lunar 0.166g)
Ch 2 A. thaliana (Lunar 0.166g)
Ch 2 A. thaliana (Earth 1.0g)
Ch 2 A. thaliana (ISS 0.0g)
B. rapa 0.166g
Ath 0.0g
Ath 1.0g
Mean Hypocotyl Length (mm)
Time After Imbibition (h)
i Tangential Root Crawling on 35 μm Nytex Screen
Ch 1/3 B. rapa (Lunar 0.166g Push)
Ch 2 A. thaliana (Lunar 0.166g Push)
Ch 2 Ath (Stagnant Pump-Off)
Ch 2 Ath (40% Foam Fill)
Tangential Root Length on Nytex (mm)
Time After Imbibition (h)